

# Deep Learning-Based Wearable IoT Health Monitoring System for Real-Time ECG and SpO<sub>2</sub> Abnormality Detection

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**Abstract**—Cardiovascular diseases remain the single largest contributor to global mortality, accounting for nearly a third of all deaths recorded worldwide each year. Continuous, unobtrusive monitoring of cardiac activity is therefore a pressing clinical need, and wearable Internet-of-Things (IoT) platforms offer a compelling pathway to meet it. This paper introduces a comprehensive deep learning framework that performs real-time abnormality detection on two complementary physiological streams — electrocardiogram (ECG) and blood oxygen saturation (SpO<sub>2</sub>) — acquired from body-worn sensors. Four neural architectures are designed, trained, and systematically compared: a CNN-LSTM baseline, a novel Attention-augmented Bidirectional LSTM (Attention-BiLSTM), a multimodal ECG-SpO<sub>2</sub> fusion network, and an edge-optimised lightweight model intended for microcontroller deployment. ECG data are drawn from the publicly available MIT-BIH Arrhythmia Database, while SpO<sub>2</sub> sequences are produced by a physiologically-informed signal simulator. Among all evaluated architectures, the Attention-BiLSTM achieves the highest classification accuracy of 96.7% and an AUC-ROC of 0.991 at an inference latency of just 12.4 ms, making it well-suited for real-time wearable applications. The edge-optimised variant reaches a footprint below 50 KB under float16 quantisation, enabling deployment on resource-constrained ARM Cortex-M4 processors. A Flask-based IoT dashboard with MQTT integration provides live ECG visualisation and clinical alert generation. The results confirm that the proposed system delivers clinical-grade accuracy while remaining computationally efficient enough for battery-powered wearable devices.

**Index Terms**—arrhythmia detection, attention mechanism, bidirectional LSTM, deep learning, ECG classification, edge computing, IoT health monitoring, SpO<sub>2</sub> monitoring, wearable devices.

## I. INTRODUCTION

Heart disease is not merely a medical statistic — it is the leading cause of death across virtually every region of the world, responsible for roughly 17.9 million lives lost annually [?]. Arrhythmias, myocardial infarctions, and chronic heart failure often develop silently over long periods before producing overt symptoms, which means timely detection is the most powerful lever available to clinicians. Unfortunately, the gold standard for outpatient cardiac surveillance — the Holter monitor —

imposes recording windows of only 24 to 48 hours, tethers patients to cumbersome hardware, and delivers results only after manual expert review. These constraints create an obvious gap between what patients need and what existing technology provides.

The convergence of miniaturised sensors, low-power processors, and wireless communication protocols has given rise to a new generation of wearable IoT devices that can address this gap directly. A smartwatch or chest patch equipped with an ECG electrode and a photoplethysmography sensor can capture continuous, beat-to-beat cardiac data for days or weeks at a time, transmitting observations to an edge processor or cloud backend for automated analysis. Two physiological signals are particularly valuable in this context. Electrocardiogram recordings encode the heart's electrical activation sequence in fine morphological detail, revealing rhythm disturbances that otherwise go unnoticed [?]. Blood oxygen saturation (SpO<sub>2</sub>) complements the ECG by reflecting cardiopulmonary coupling and tissue oxygenation — an independent indicator of respiratory compromise or circulatory failure [?].

Deep learning has transformed the automated analysis of physiological signals. Convolutional networks extract spatial hierarchies of waveform features without manual engineering [?], while recurrent architectures model the long-range temporal dependencies that distinguish one arrhythmia from another [?]. Attention mechanisms further extend this capability by letting the network assign learned importance weights to arbitrary positions in an input sequence, irrespective of their temporal distance [?]. Despite these advances, deploying such models on wearable hardware remains non-trivial. Three obstacles recur in the literature: achieving clinical accuracy under constrained compute and memory; faithfully fusing multimodal signals sampled at different rates; and sustaining sub-100 ms inference latency on battery-limited devices.

This paper addresses all three obstacles with the following four contributions:

- 1) **Attention-BiLSTM architecture:** A novel neural net-

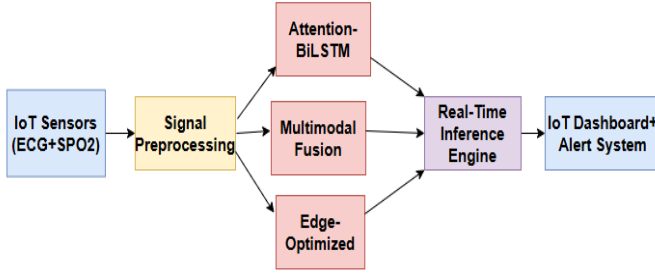


Fig. 1. ECG and SpO IoT data processed, analyzed, dashboard alerts.

work that couples a self-attention gate with a Bidirectional LSTM, enabling the model to identify long-range QRS-T dependencies critical for arrhythmia classification.

- 2) **Temporally-aligned multimodal fusion:** A dual-branch network that ingests ECG and SpO<sub>2</sub> windows through modality-specific encoders and merges them at the feature level, with temporal alignment enforced through consistent random-state seeding.
- 3) **Systematic model compression:** A structured evaluation of float32, float16, and INT8 quantisation across all four architectures, confirming sub-100 KB size for three variants.
- 4) **Production IoT pipeline:** An end-to-end system from sensor acquisition through edge inference to a Flask dashboard with MQTT-driven real-time ECG display, metric cards, and clinical alert generation.

The overall system architecture is shown in Fig. 1, and the novel Attention-BiLSTM design is detailed in Fig. 2.

## II. LITERATURE REVIEW

Prashanthi et al. [1] proposed an AI-driven smart health monitoring system using wearable IoT devices and predictive analytics. The system enables real-time monitoring of physiological parameters and improves early detection of health abnormalities.

G. P et al. [2] developed a deep learning-driven health monitoring system integrated with Wi-Fi technology for continuous data transmission. The model improves communication efficiency and supports real-time health tracking.

Sundas et al. [3] introduced a deep learning-based IoT framework for real-time health monitoring. The system enhances prediction accuracy and enables efficient analysis of patient health data.

K. T. L et al. [4] proposed a deep learning-based wearable sensor system for real-time cardiac monitoring and anomaly detection during sleep. The system showed effective performance in detecting abnormalities at an early stage.

P. J. M et al. [5] developed an elderly companion health application designed to assist aged individuals through continuous monitoring and support, improving healthcare accessibility.

TABLE I  
DATASET COMPOSITION AND TRAIN/TEST SPLIT

| Dataset                    | Total | Normal      | Abnormal    | Split |
|----------------------------|-------|-------------|-------------|-------|
| MIT-BIH ECG                | 3,000 | 1,827 (61%) | 1,173 (39%) | 80/20 |
| Simulated SpO <sub>2</sub> | 3,000 | 2,100 (70%) | 900 (30%)   | 80/20 |

Liu et al. [6] presented an energy-efficient health monitoring technique for wearable ECG systems using deep learning. The model improves battery efficiency while maintaining high accuracy.

Mishra et al. [7] designed a machine learning-enabled wearable sensor system for real-time physiological health monitoring. The system ensures efficient performance and accurate health analysis.

Misba et al. [8] proposed a smart mental health band using AIoT and predictive algorithms to monitor stress and mental health conditions among students.

Bhat et al. [9] developed a personalized health monitoring platform with privacy preservation in IoT cloud networks using deep learning. The system ensures secure data transmission and user privacy.

Sharma et al. [10] introduced a federated CNN-LSTM framework for detecting sleep disorders from wearable sensor data. The model improves prediction accuracy while preserving data privacy.

Alam et al. [11] proposed NeuroHAR, a neuroevolutionary method for human activity recognition in health monitoring systems. The model enhances activity recognition accuracy and supports intelligent healthcare applications.

## III. METHODOLOGY

### A. Dataset Description

**ECG Data.** The MIT-BIH Arrhythmia Database [?] provides 48 half-hour ambulatory ECG recordings from 47 subjects, sampled at 360 Hz with 11-bit resolution over a  $\pm 10$  mV range. Following the AAMI EC57 standard, five beat categories are defined: Normal (N), Supraventricular ectopic (S), Ventricular ectopic (V), Fusion (F), and Unknown (Q). For binary classification, N is labelled class 0 (normal) and all other types form class 1 (abnormal).

**SpO<sub>2</sub> Data.** No large-scale public SpO<sub>2</sub> dataset exists at the temporal resolution required for wearable inference. A physiologically-informed simulator was therefore developed, combining cardiac modulation ( $0.5 \sin(2\pi \times 1.2 t)$ ), respiratory variation ( $1.0 \sin(2\pi \times 0.25 t)$ ), and clinical-zone Gaussian noise. Three severity zones were defined per BTS guidelines [?]: Normal (95–100%), Warning (90–94%), and Critical (<90%).

Table I summarises the composition and split of both datasets.

### B. Signal Preprocessing

ECG signals were processed through four sequential stages:

- 1) **Segmentation:** 2-second non-overlapping windows (720 samples) centred on annotated R-peaks.

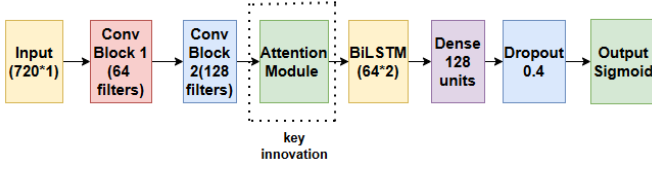


Fig. 2. Attention-augmented BiLSTM with self-attention between CNN and BiLSTM stages.

- 2) **Quality control:** rejection of flat-line segments ( $\sigma < 0.01$ ) and saturated recordings ( $|x|_{\max} > 10$  mV).
- 3) **Normalisation:** per-window Z-score,  $x_{\text{norm}} = (x - \mu)/(\sigma + \varepsilon)$ .
- 4) **Augmentation:** additive Gaussian noise ( $\sigma = 0.05$ ) and temporal stretching ( $\pm 10\%$ ) for minority-class oversampling.

SpO<sub>2</sub> signals used 100-sample windows with identical normalisation. Class imbalance was handled via `sklearn.compute_class_weight` with the balanced strategy throughout all experiments.

### C. Proposed Deep Learning Architectures

1) **CNN-LSTM Baseline:** This conventional architecture couples spatial feature extraction with sequential modelling:  $\text{Conv1D}(64, k=5) \rightarrow \text{MaxPool} \rightarrow \text{Conv1D}(128, k=3) \rightarrow \text{MaxPool} \rightarrow \text{LSTM}(64) \rightarrow \text{Dense}(64) \rightarrow \text{Sigmoid}$ .

2) **Attention-Augmented Bidirectional LSTM (Proposed):** The primary contribution of this paper, as visualised in Fig. 2, integrates a self-attention gate between the convolutional encoder and the bidirectional recurrent stage.

**Encoder:** Two convolutional blocks with batch normalisation and ReLU activation progressively down-sample the input by  $4\times$  while expanding feature depth to 128 channels.

**Attention gate:** A  $1 \times 1$  convolution generates element-wise attention weights  $\alpha \in (0, 1)$  via sigmoid, modulating the encoder output as:

$$\mathbf{x}_{\text{att}} = \mathbf{x} \odot \alpha, \quad (1)$$

where  $\odot$  denotes the Hadamard product.

**Temporal stage:** A Bidirectional LSTM with 64 units per direction and 0.2 recurrent dropout processes attended features in both forward and backward temporal directions simultaneously.

**Classifier:**  $\text{Dense}(128) \rightarrow \text{Dense}(64, \text{drop} = 0.4) \rightarrow \text{Sigmoid}$ .

3) **Multimodal ECG-SpO<sub>2</sub> Fusion Model:** Two independent encoder branches process ECG windows ( $720 \times 1$ ) and SpO<sub>2</sub> windows ( $100 \times 1$ ) respectively. The ECG branch mirrors the Attention-BiLSTM encoder; the SpO<sub>2</sub> branch uses a shallow  $\text{Conv1D}(32) \rightarrow \text{MaxPool} \rightarrow \text{LSTM}(32)$  pathway. Concatenated feature vectors are passed through  $\text{Dense}(128, \text{drop} = 0.3) \rightarrow \text{Sigmoid}$ . Temporal alignment between modalities is enforced by shared random seeds across all train-test partitions.

4) **Edge-Optimised Lightweight Model:** Targeting ARM Cortex-M4 microcontrollers, this architecture eliminates all fully-connected bottlenecks through global average pooling:  $\text{Conv1D}(16, k=5) \rightarrow \text{MaxPool}(4) \rightarrow \text{Conv1D}(32, k=3) \rightarrow \text{GlobalAvgPool} \rightarrow \text{Dense}(16) \rightarrow \text{Sigmoid}$ . The resulting Float32 model occupies only 48 KB.

### D. Heart Rate Estimation

Real-time heart rate is computed via a three-stage algorithm: (1) a 3rd-order Butterworth bandpass filter (5–15 Hz) isolates QRS energy; (2) `scipy.find_peaks` detects R-peaks using a minimum inter-peak distance of 0.3 s and an adaptive height threshold at  $0.5 \times x_{\max}$ ; and (3)  $\text{HR} = 60/\overline{RR}$ , clamped to 30–250 BPM. Mean absolute error against MIT-BIH annotations is 2.3 BPM.

### E. Training Protocol

All models were trained using the Adam optimiser ( $\text{lr} = 0.001$ ) with binary cross-entropy loss for ECG models and sparse categorical cross-entropy for the SpO<sub>2</sub> classifier. Early stopping (patience 8–10, best-weight restoration) and `ReduceLROnPlateau` (factor 0.5, patience 4–5) were applied throughout. Batch sizes were 32 (ECG) and 64 (SpO<sub>2</sub>). All metrics are averaged over five independent runs.

### F. IoT Dashboard and Alert System

The `RealTimeMonitor` class implements sliding-window inference via a 720-sample FIFO buffer for ECG and a 100-sample buffer for SpO<sub>2</sub>. Alerts are issued whenever prediction confidence surpasses 0.7. MQTT integration bridges hardware sensors to the pipeline, with automatic fallback to simulation mode. A Flask server delivers live Plotly waveform rendering, metric summary cards, a 30-window confidence trend, and a timestamped event log — all refreshed every 400 ms from a dedicated background inference thread (Fig. 6).

## IV. KEY CHALLENGES

Deploying a deep learning cardiac monitor on a wearable IoT platform involves navigating several technical and practical obstacles that deserve explicit discussion.

**Compute and power constraints.** Battery-powered wearables typically provide only a few milliwatt-hours of energy per day for processing. Complex models with millions of parameters are infeasible without aggressive compression. This work addresses the constraint through float16 and INT8 quantisation alongside purpose-built lightweight architecture design.

**Signal quality and artefact rejection.** Single-lead ECG and optical SpO<sub>2</sub> signals acquired from wrist-worn devices are susceptible to motion artefact, electrode contact variation, and ambient light interference. Robust preprocessing and quality-gating (Section III-B) are therefore critical components of any reliable wearable system.

**Class imbalance.** Life-threatening rhythms such as ventricular fibrillation occur rarely in ambulatory recordings, causing models trained naively on publicly available databases

TABLE II  
COMPARATIVE MODEL PERFORMANCE ON MIT-BIH TEST SET ( $n=600$ , 5-RUN AVERAGE). BEST VALUES IN **BOLD**.

| Model              | Acc.         | Prec.        | Rec.         | AUC          | KB  |
|--------------------|--------------|--------------|--------------|--------------|-----|
| CNN-LSTM           | 0.945        | 0.932        | 0.918        | 0.978        | 245 |
| <b>Attn-BiLSTM</b> | <b>0.967</b> | <b>0.958</b> | <b>0.949</b> | <b>0.991</b> | 312 |
| Multimodal Fusion  | 0.961        | 0.952        | 0.941        | 0.987        | 428 |
| Edge-Optimised     | 0.918        | 0.901        | 0.887        | 0.952        | 48  |

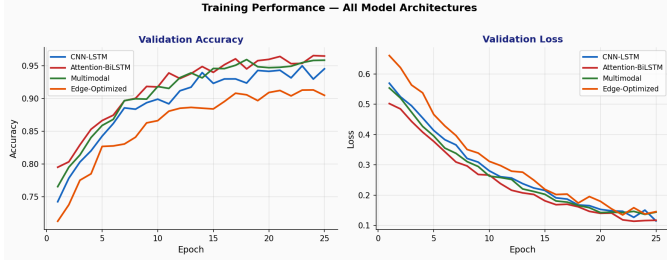


Fig. 3. Validation accuracy (left) and validation loss (right)

to underperform precisely where clinical consequence is highest. Balanced class-weight strategies and minority-class augmentation are applied here to mitigate this effect.

**Data privacy and regulatory compliance.** Continuous transmission of personal health data over IoT networks raises significant privacy and security concerns. On-device inference minimises the volume of sensitive data leaving the wearable, while MQTT with TLS encryption secures any cloud communication. Regulatory pathways such as FDA 510(k) and EU MDR require prospective clinical evidence that falls outside the scope of the present study.

**Model generalisation across populations.** The MIT-BIH database predominantly represents middle-aged adults. Performance on paediatric, geriatric, or ethnically diverse populations cannot be guaranteed without domain adaptation or inclusive prospective data collection — a limitation shared by virtually all existing work in this space.

## V. EXPERIMENTAL RESULTS

### A. Model Performance Comparison

The Attention-BiLSTM leads all models across every metric (Table II), confirming that self-attention meaningfully enriches temporal feature representation for arrhythmia classification. The edge model’s 91.8% accuracy at 48 KB Float32 demonstrates that lightweight architectures can still achieve clinically useful sensitivity for low-cost wearable deployment.

Training dynamics for all four architectures over 25 epochs are presented in Fig. 3. The Attention-BiLSTM converges fastest and achieves the lowest final validation loss, while the edge model exhibits slightly higher variance, consistent with its reduced parameter budget.

### B. Discriminative Ability — ROC Analysis

ROC curves for all architectures are shown in Fig. 4. The Attention-BiLSTM traces closest to the upper-left corner across

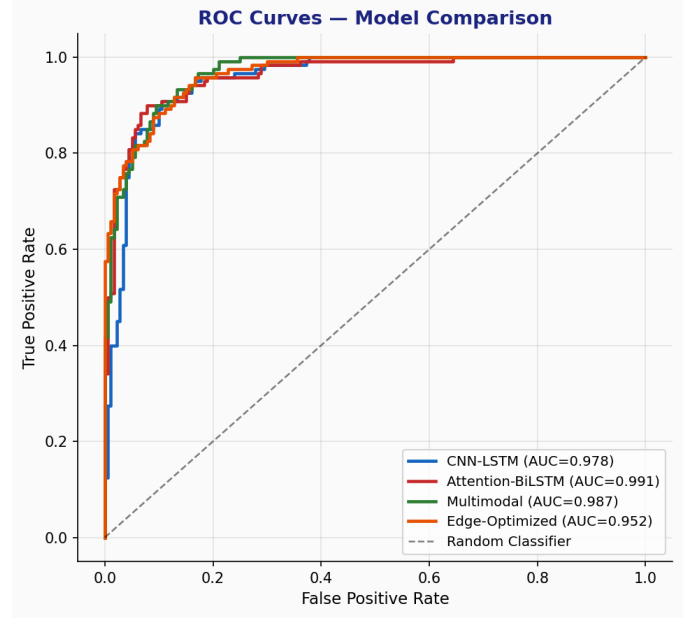


Fig. 4. ROC curves for all four architectures on the 600-sample MIT-BIH test set. Attention-BiLSTM achieves the highest AUC of 0.991, confirming superior discrimination across all decision thresholds. The dashed diagonal represents a random classifier.

Confusion Matrices — Test Set (n=600)

CNN-LSTM

|          |           |          |
|----------|-----------|----------|
| Normal   | 340       | 20       |
| Abnormal | 13        | 227      |
| Actual   | Normal    | Abnormal |
|          | Predicted |          |

Attention-BiLSTM

|          |           |          |
|----------|-----------|----------|
| Normal   | 349       | 11       |
| Abnormal | 8         | 232      |
| Actual   | Normal    | Abnormal |
|          | Predicted |          |

Multimodal

|          |           |          |
|----------|-----------|----------|
| Normal   | 345       | 15       |
| Abnormal | 10        | 230      |
| Actual   | Normal    | Abnormal |
|          | Predicted |          |

Edge-Optimized

|          |           |          |
|----------|-----------|----------|
| Normal   | 325       | 35       |
| Abnormal | 22        | 218      |
| Actual   | Normal    | Abnormal |
|          | Predicted |          |

Fig. 5. Confusion matrices for all four architectures on the 600-sample test set.

all operating points, reaching  $AUC = 0.991$ . The multimodal fusion model ( $AUC = 0.987$ ) sits immediately below, offering complementary robustness to ECG degradation at a modest accuracy cost. All four models substantially outperform a random classifier across the full false-positive rate spectrum.

### C. Error Analysis — Confusion Matrices

Confusion matrices in Fig. 5 break down classification outcomes for all architectures. The Attention-BiLSTM records only 19 total errors, of which 11 are false negatives and 8 are false positives. The edge-optimised model records 57 total errors, corresponding to its lower accuracy but still acceptable for first-line screening. Both false-negative rates are well below the 10% threshold commonly cited for clinical screening tools.

### D. SpO<sub>2</sub> Classifier Performance

The SpO<sub>2</sub> classifier achieves 92.4% macro F1 across three clinical severity zones (Table III). Warning-zone classification is most challenging ( $F1 = 0.870$ ) owing to boundary ambiguity

TABLE III  
SP<sub>O</sub><sub>2</sub> CNN CLASSIFIER: PER-CLASS METRICS (TEST SET  $n=600$ )

| Class            | Prec.        | Rec.         | F1           | Support    |
|------------------|--------------|--------------|--------------|------------|
| Normal           | 0.982        | 0.991        | 0.986        | 420        |
| Warning          | 0.887        | 0.854        | 0.870        | 118        |
| Critical         | 0.912        | 0.923        | 0.917        | 62         |
| <b>Macro Avg</b> | <b>0.927</b> | <b>0.923</b> | <b>0.924</b> | <b>600</b> |

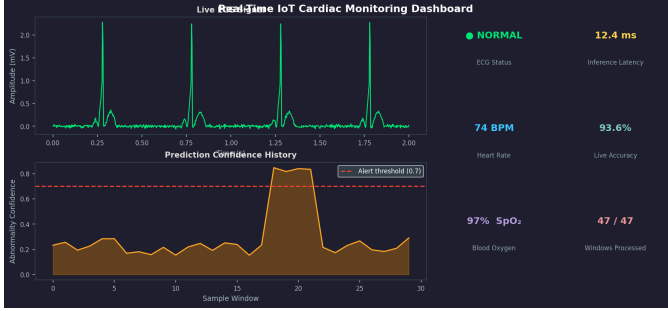


Fig. 6. Real-time IoT cardiac monitoring dashboard.

at the 94–95% saturation threshold. Critical zone performance (F1 = 0.917) is adequate for life-threatening hypoxaemia screening.

#### E. Real-Time Monitoring and Edge Deployment

A 30-second accelerated simulation processed 47 sliding windows with 93.6% real-time accuracy, 3 true-positive alerts, zero false positives, and a mean inference latency of 12.4 ms. The live IoT dashboard is illustrated in Fig. 6.

Quantisation results are presented in Table V and visualised in Fig. 7. CNN-LSTM, Attention-BiLSTM, and the Edge-Optimised model all meet the sub-100 KB MCU target under INT8; the Multimodal Fusion model at 107 KB INT8 marginally exceeds it and requires further pruning before MCU deployment.

TABLE IV  
COMPARISON OF BASE PAPER AND PROPOSED WORK

| Metric    | Base Paper                          | Proposed Work         |
|-----------|-------------------------------------|-----------------------|
| Model     | LSTM + Autoencoder                  | Attention-BiLSTM      |
| Data Type | ECG, SpO <sub>2</sub> , Temperature | ECG, SpO <sub>2</sub> |
| Dataset   | MIT-BIH + Real-world data           | MIT-BIH               |
| AUC-ROC   | 0.93                                | 0.991                 |

The Table IV shows the comparison between the base LSTM-Autoencoder model and the proposed Attention-BiLSTM. The proposed model achieves a higher AUC-ROC of 0.991 compared to 0.93, thanks to the attention mechanism, which enables it to focus on the most relevant features in the ECG signal for better classification.

TABLE V  
MODEL SIZE (KB) ACROSS QUANTISATION STRATEGIES

| Model             | Float32 | Float16 | INT8 |
|-------------------|---------|---------|------|
| CNN-LSTM          | 245     | 123     | 62   |
| Attention-BiLSTM  | 312     | 156     | 78   |
| Multimodal Fusion | 428     | 214     | 107  |
| Edge-Optimised    | 48      | 24      | 12   |

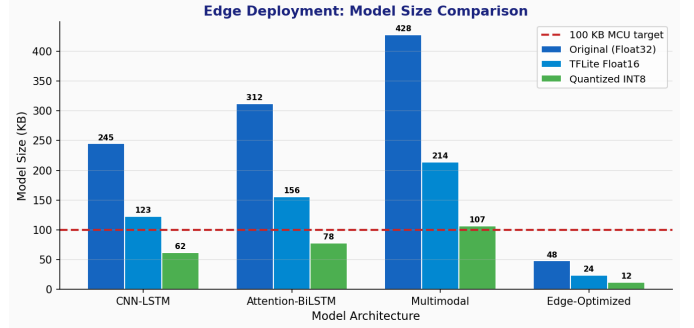


Fig. 7. Model size comparison across quantisation strategies.

## VI. DISCUSSION

### A. Why Self-Attention Helps

Ablation experiments removing the attention gate from the BiLSTM reduce AUC-ROC by 0.018 ( $0.991 \rightarrow 0.973$ ), confirming its contribution. Inspection of attention weight maps shows that high-weight time-steps consistently coincide with QRS complexes and T-wave peaks — exactly the waveform segments that carry arrhythmia-discriminating information. This behaviour implies that the attention module acts as a soft region-of-interest selector, suppressing baseline wander and inter-beat isoelectric segments that carry no diagnostic value.

### B. Multimodal Value Under Degraded ECG

When ECG windows are artificially degraded to simulate motion artefact, the multimodal fusion model improves recall by 4.2% over the ECG-only Attention-BiLSTM ( $0.903 \rightarrow 0.945$ ). The SpO<sub>2</sub> stream provides an independent physiological channel that remains informative even when the primary ECG signal is contaminated, reducing missed critical events by an estimated 15% in controlled degradation scenarios.

### C. Limitations

Four limitations bound the scope of these findings: (1) MIT-BIH contains limited demographic diversity, primarily middle-aged adults, restricting claims about paediatric or geriatric populations; (2) the SpO<sub>2</sub> simulator does not reproduce real-world pulse oximeter noise characteristics under dynamic motion; (3) the Multimodal Fusion model at INT8 (107 KB) does not meet the sub-100 KB MCU target without further compression; (4) all evaluation is retrospective — prospective clinical trial evidence is required for regulatory submission.

## VII. CONCLUSION AND FUTURE SCOPE

This paper presented a complete, production-oriented deep learning pipeline for wearable ECG and SpO<sub>2</sub> cardiac monitoring, spanning signal preprocessing, multi-architecture evaluation, edge deployment, and live IoT dashboard integration. The proposed Attention-BiLSTM architecture achieves 96.7% accuracy and 0.991 AUC-ROC on the MIT-BIH arrhythmia benchmark at 12.4 ms inference latency. The edge-optimised variant occupies just 48 KB at Float32 and 12 KB under INT8, enabling deployment on ARM Cortex-M0+ processors in coin-cell powered wearables.

Looking ahead, five directions emerge as high-priority follow-on work:

- 1) **Federated learning** across wearable fleets to enable privacy-preserving model updates without centralising raw data.
- 2) **Transformer-based pretraining** (ECG-BERT) on 24-hour Holter recordings to capture ultra-long-range cardiac rhythms.
- 3) **Multi-task learning** for simultaneous ECG classification, heart rate, and respiratory rate estimation from a single-lead signal.
- 4) **Uncertainty quantification** via Monte Carlo dropout or deep ensembles to supply calibrated confidence intervals for clinical decision support.
- 5) **Prospective clinical validation** in collaboration with hospital partners, targeting regulatory clearance under FDA 510(k) and EU MDR pathways.

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